Audio Feature Extraction Documentation

Overview

This system extracts numerical features from audio signals to convert raw sound into a structured format suitable for machine learning. Each audio file is transformed into a fixed-length feature vector of 220 values.

Feature Categories

1. MFCC (Mel-Frequency Cepstral Coefficients) – 80 Features

- 40 Mean values
- 40 Standard Deviation values

Description:
MFCC represents the short-term power spectrum of sound using a perceptual scale (Mel scale), which mimics human hearing.

Purpose:

- Captures the overall shape and timbre of audio
- Essential for distinguishing between normal and faulty machine sounds

2. Delta MFCC – 80 Features

- 40 Mean values
- 40 Standard Deviation values

Description:
Delta MFCC measures the rate of change of MFCC features over time.

Purpose:

- Captures temporal dynamics
- Helps detect instability and irregular patterns in sound

3. Spectral Features – 10 Features

Each feature includes mean and standard deviation:

- Spectral Centroid
- Spectral Bandwidth
- Spectral Rolloff
- Zero Crossing Rate (ZCR)

- Root Mean Square Energy (RMS)

Description & Purpose:

- Spectral Centroid: Indicates brightness of sound
- Spectral Bandwidth: Measures spread of frequencies
- Spectral Rolloff: Identifies high-frequency energy distribution
- ZCR: Measures signal noisiness
- RMS: Represents signal strength (loudness)

4. Spectral Contrast – 14 Features

- 7 Mean values
- 7 Standard Deviation values

Description:
Measures the difference between peaks and valleys in the frequency spectrum.

Purpose:

- Distinguishes structured signals from noisy signals
- Useful for identifying degradation in machine sound

5. Chroma Features – 24 Features

- 12 Mean values
- 12 Standard Deviation values

Description:
Represents energy distribution across 12 pitch classes.

Purpose:

- Captures repeating patterns in frequency
- Useful for periodic or rotating machine components

6. Tonnetz Features – 12 Features

- 6 Mean values
- 6 Standard Deviation values

Description:
Represents harmonic relationships in audio.

Purpose:

- Captures deeper harmonic structure

- Helps identify inconsistencies in periodic signals

Total Feature Count

Total features extracted per audio file:

- MFCC: 80
- Delta MFCC: 80
- Spectral Features: 10
- Spectral Contrast: 14
- Chroma: 24
- Tonnetz: 12

Total = 220 Features

Workflow Summary

1. Load audio file (resampled to 22050 Hz)
2. Extract multiple feature types
3. Compute mean and standard deviation for each feature
4. Combine all features into a single vector
5. Use the vector as input for machine learning models

Key Insight

The model does not process raw audio directly. Instead, it learns patterns from the extracted numerical features, enabling it to classify audio as normal or faulty based on learned characteristics.

Application

This feature extraction pipeline is suitable for:

- Machine fault detection
- Audio classification tasks
- Predictive maintenance systems